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HELICAL SEAL FOR MULTI-STAGE COMPRESSOR

Abstract

A compressor comprising: an drive shaft; a first compressor wheel; a second compressor wheel connected to the first compressor wheel via the drive shaft; and two or more helical seals disposed on the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the drive shaft and around a circumference of the drive shaft, wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel.

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Background/Summary

[0001] This application claims the benefit of Indian Provisional Patent Application No. 20/241,101/7346, filed Mar. 11, 2024 and entitled “HELICAL SEAL FOR MULTI-STAGE COMPRESSOR,” the entire contents of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to systems and techniques for producing conditioned air for a vehicle, and more particularly, to compressor systems and techniques for producing compressed vapor refrigerant.

BACKGROUND

[0003] A vapor cycle system may cool a fresh air stream using a refrigerant. The vapor cooling system compresses and condenses the refrigerant from a relatively low-pressure vapor to a relatively high-pressure liquid, which then expands and evaporates to remove heat from the fresh air stream.

SUMMARY

[0004] A vapor cycle system (VCS) may monitor and control the temperatures of systems and/or components within a vehicle. VCS may include one or more centrifugal compressors which compresses and condenses a refrigerant flowing within the VCS from a relative-low pressure vapor to a relatively high-pressure fluid. In some examples, a compressor system of a VCS may include multiple centrifugal compressors coupled in series to cause the compressor system to output the refrigerant at a specific pressure ratio (e.g., a ratio of the pressure of the refrigerant at an output of the compressor system to the pressure of the refrigerant at an input of the compressor system). When multiple compressors are coupled in series, each compressor may receive the refrigerant from an outlet of a preceding compressor in the series and output the refrigerant directly into a succeeding compressor in the series via an outlet of the compressor.

[0005] Compressor systems may multiple compressors coupled in series and connected via an drive shaft extending through each compressor. In other compressor systems, there may be unintended transfer of refrigerant between compressors, e.g., along a channel within the compressor system configured to retain the drive shaft. The example compressor systems described herein include a plurality of helical seals disposed on the drive shaft at a plurality of locations along the drive shaft. The helical seals may inhibit the flow of refrigerant through the channel and between longitudinally adjacent compressors.

[0006] The example compressor system with multiple helical seals may provide several advantages over other compressor systems with multiple compressors or compressor wheels coupled in series. The example helical seals described herein may facilitate the redirection of refrigerant within the channel and inhibit the flow of refrigerant past each helical seal and towards another compressor. In some examples, the example helical seals may be wound in different directions, which may facilitate re-direction of proximal and distal flow of refrigerant out of the channel, which may inhibit or reduce the unintended flow of refrigerant from a distal compressor into a proximal compressor, and vice versa. The example helical seals may provide increased efficiency, simplicity, and modularity compared to other seals (e.g., carbon seals) disposed within other compressor systems with multiple compressors. The example helical seals may facilitate balancing of the drive shaft within the channel while inhibiting the unintended flow of refrigerant.

[0007] In some examples, the disclosure is directed to a compressor comprising: an drive shaft; a first compressor wheel; a second compressor wheel connected to the first compressor wheel via the drive shaft; and two or more helical seals disposed on the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the drive shaft and around a circumference of the drive shaft, wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel.

[0008] In some examples, the disclosure is directed to a compressor system comprising: a compressor comprising: a first compressor wheel; a second compressor wheel; an drive shaft

extending along a longitudinal axis from a proximal end to a distal end, wherein the drive shaft extends through the first compressor wheel and the second compressor wheel and connects the first compressor wheel to the second compressor wheel, and wherein the distal end of the drive shaft is distal to the first compressor wheel; and two or more helical seals disposed on the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the drive shaft and around a circumference of the drive shaft; a housing configured to retain the compressor, the housing comprising: a channel extending along the longitudinal axis and configured to retain the drive shaft, wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel through the channel; and a motor coupled to the proximal end of the drive shaft and configured to apply a rotary force on the drive shaft.

[0009] In some examples, the disclosure is directed to a method comprising: position a first compressor wheel of a compressor within a housing; connecting a second compressor wheel of the compressor to the first compressor wheel by at least partially inserting an outer shaft of an drive shaft of the compressor into the first compressor wheel, wherein the outer shaft is connected to the second compressor wheel, and wherein the outer shaft of the drive shaft comprises: two or more helical seals disposed on the outer shaft, each helical seal of the two or more helical seals extending away from an outer surface of the outer shaft and around a circumference of the outer shaft; inserting an inner shaft of the drive shaft through an inner lumen of the outer shaft along a longitudinal axis of the compressor; and affixing the inner shaft to the first compressor wheel and the second compressor wheel.

[0010] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

[0012] FIG. 1 is a block diagram illustrating a system for generating conditioned air that includes an example vapor cooling system (VCS).

[0013] FIG. 2 is a conceptual diagram illustrating an example vapor cycle compressor (VCC).

[0014] FIG. 3 is a conceptual diagram illustrating an example second-stage compressor wheel and example drive shaft of the example centrifugal compressor of FIG. 2.

[0015] FIG. 4A is a conceptual diagram illustrating an example of the outer shaft of FIG. 3.

[0016] FIG. 4B is a conceptual diagram illustrating another view of the example outer shaft of FIG. 4A.

[0017] FIG. 5 is a flow diagram illustrating an example process of assembling the VCC of FIG. 2.

[0018] Like reference characters refer to like elements throughout the figures and description.

DETAILED DESCRIPTION

[0019] Various examples discussed herein describe vapor cooling systems, vapor compression systems, centrifugal compressors, centrifugal compressor systems, and centrifugal compressor components that include two or more compressor wheels coupled in series and in a nested configuration.

[0020] A centrifugal compressor system is configured to drive one or more impellers of one or more compressor wheels using a compressor motor to compress a vapor refrigerant to a higher pressure. Compressor wheels may be coupled in series to increase a pressure ratio generated by the

compressor system. In such examples, a first compressor wheel may compress a refrigerant and may output the refrigerant directly into a second compressor wheel. The second compressor wheel may then further compress the refrigerant, thereby increasing the pressure ratio of the refrigerant. Multiple impellers of multiple compressor wheels coupled in series may be driven by a same compressor motor.

[0021] In some examples, a centrifugal compressor system with multiple compressor wheels coupled in series may position multiple compressor wheels along a longitudinal axis. For example, a first-stage compressor wheel may be distal to a second-stage compressor wheel. Each compressor wheel may be connected (e.g., to a motor) via an drive shaft extending through the compressor wheels. In such examples, the first-stage compressor wheel may be affixed to a distal end or distal portion of the drive shaft and the second-stage compressor wheel may be affixed to the drive shaft at a position proximal to the first-stage compressor wheel. Each compressor wheel may be separated from one or more other compressor wheels via the housing of the compressor system, e.g., such that refrigerant is configured to be directed from outlet of one compressor wheel to an inlet of another compressor wheel via one or more pathways defined by the housing.

[0022] In some multi-stage compressor systems, refrigerant may flow along a channel in the housing configured to retain the drive shaft. In some examples, the refrigerant may flow along the channel and past one or more compressor wheels (e.g., past the first-stage compressor wheel and directly into the second-stage compressor wheel). In some examples, the refrigerant may flow from a compressor wheel directly towards an inlet or outlet of the compressor system (e.g., from second-stage compressor wheel and towards an inlet of the first-stage compressor wheel).

[0023] The disclosure describes a compressor system with two or more compressor wheels connected in series via an drive shaft. The compressor system may include two or more helical seals disposed on the drive shaft (e.g., directly on an outer surface of the drive shaft, on an outer surface of an outer casing configured to be secured to the drive shaft). The two or more helical seals may be configured to inhibit flow of refrigerant through the channel containing the drive shaft. Each helical seal may be wound in a different direction than another helical seal. The helical seals may be wound in specific directions to redirect the flow of refrigerant through the channel, e.g., back towards an entrance to the channel through which the refrigerant enters the channel.

[0024] The compressor system described herein may provide advantages over other compressor systems. The compressor system described here may include helical seals which provide improved sealing of the channel around the drive shaft without impeding the rotation of the drive shaft within the channel. The helical seals may be integrated into the drive shaft and/or an outer casing of the drive shaft, which may simplify maintenance and replacement of the helical seals (e.g., via simple replacement of the drive shaft and/or one or more compressor wheels) compared to other seals (e.g., carbon seals). The helical seals may provide improved abilities to redirect the flow of refrigerant within the channel, which may further improve the sealing capabilities of the helical seals. The compressor system described herein may be less complex and less costly to manufacture than other compressor systems due to the reduced complexity of the housing of the compressor system and/or the reduced complexity of the arrangement of the compressor wheels.

[0025] Vapor cooling systems, compressors, compressor systems, and compressor components discussed herein may be used to produce conditioned air for a variety of applications. In some examples, vapor cooling systems discussed herein may be used to cool pressurized air, such as for a pressurized cabin or avionics systems of an aircraft. In some examples, vapor cooling systems, including vapor-cooled compression systems, may be used, for aircraft and non-aircraft implementations, to cool liquid, non-pressurized air, etc., in accordance with one or more of the various techniques of this disclosure. In another example, vapor cooling systems, including vapor-cooled compression systems, may be used, for aircraft and non-aircraft implementations, to cool equipment, such as through direct contact-cooling of equipment.

[0026] FIG. 1 is a block diagram illustrating an example system **100** for generating conditioned air

that includes a vapor cooling system (VCS) **104**. The conditioned air may be used to cool volumes or components of various cabins or avionics systems **122**. Cabin/avionics **122** may be a compartment of a vehicle (e.g., an aircraft, an automobile, a spacecraft, a watercraft, etc.) that includes an internal environment and/or one or more avionics systems that receive cooled air for cooling equipment. For example, cabin/avionics **122** may be configured to house people, cargo, and the like, in the internal environment. It will be understood that avionics generally relate to aircraft, spacecraft, etc., and that other systems may include other electronic systems/control systems configured for cooling. Thus, while described as cabin/avionics **122**, the techniques of this disclosure are not so limited, and a person skilled in the art will understand that the systems described herein may be employed in a variety of contexts without significantly departing from structures and mechanics described herein.

[0027] VCS **104** includes a vapor cycle compressor (VCC) **112**, a condenser **106**, an expansion device **108** (e.g., an expansion valve), and an evaporator **110** fluidically coupled to each other through pressurized refrigerant supply lines to form a refrigerant circuit. A variety of refrigerants may be used in VCS **104**, as will be explained further below.

[0028] VCC **112** may be configured to receive vapor refrigerant from evaporator **110** and compress and pump vapor refrigerant to condenser **106**. VCC **112** may include a centrifugal compressor system configured to receive the vapor refrigerant at an inlet pressure and discharge the vapor refrigerant at a higher outlet pressure. VCC **112** may include one or more compression stages and an electrically driven motor. The motor may be configured to receive electrical power, such as from a motor controller **120**, and generate mechanical power to drive the one or more compression stages. Condenser **106** may be configured to receive saturated vapor refrigerant from VCC **112**, condense the vapor refrigerant, and discharge saturated refrigerant to an expansion device **108**. Condenser **106** may be cooled by environmental air, such as ram air flow, from a ram air system **102**, or another fluid such as fuel or heat transport fluids.

[0029] Expansion device **108** may be configured to control flow of refrigerant to evaporator **110** and reduce a pressure of saturated refrigerant prior to entry into evaporator **110**. Expansion device **108** may be an orifice, tube, metered valve, or other device configured to reduce a pressure of a saturated refrigerant. Evaporator **110** may be configured to receive cabin pressure air, such as from an air supply system **124**, remove heat from cabin air using a refrigerant, and discharge cabin air to cabin/avionics **122**. On a refrigerant side, evaporator **110** may be configured to receive saturated refrigerant, absorb heat from the cabin air, vaporize the refrigerant, and discharge superheated vapor refrigerant.

[0030] System **100** includes a control system **114** for controlling various conditions of VCS **104**, such as refrigerant flow rate, refrigerant vapor composition, refrigerant temperature, and the like. Control system **114** may be configured to monitor and/or operate one or more process control components of system **100**. For example, control system **114** may be communicatively coupled to any of air supply system **124**, ram air system **102**, VCC **112**, expansion device **108**, or any other component of system **100**. Control system **114** may also be communicatively coupled to instrumentation, such as flow meters, temperature sensors, and pressure sensors, and configured to receive measurement signals from the instrumentation. For example, control system **114** may be configured to receive measurement signals for various parameters of VCS **104**, such as a speed of VCC **112**, temperature of cabin air leaving evaporator **110**, or a superheat of vapor refrigerant entering VCC **112**, determine a mismatch between the measurement signals and a setpoint for the corresponding parameter, and send a control signal to one or more components of system **100** to reduce the mismatch and return the parameter to within the setpoint. Control system **114** may include any of a wide range of devices, including processors (e.g., one or more microprocessors, one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), or the like), servers, desktop computers, notebook (i.e., laptop) computers, tablet computers, and the like.

[0031] Control system **114** may include a motor controller **120** configured to control a motor of VCC **112**. As discussed above, a motor may provide mechanical power to impellers of VCC **112**, and therefore modulate flow rate of refrigerant through VCS **104**. The speed of VCC **112** may correspond to various temperature setpoints of VCS **104**, such as cooling demands of evaporator **110** and inlet superheat of the vapor refrigerant into VCC **112**. To control a speed of VCC **112**, motor controller **120** may be configured to send control signals to VCC **112** to control an amount of electrical power to the motor of VCC **112**, such as from an APU or other power supply. Control system **114** may include a vapor controller **116** configured to control a vapor composition of the refrigerant in VCS **104**. To control a vapor composition of the refrigerant, vapor controller **116** may be configured to send control signals to expansion device **108** and/or evaporator **110** to control a position of expansion device **108** and/or a position of a bypass valve of evaporator **110**, such as by closing expansion device **108** to increase a superheat of the refrigerant entering VCC **112**. Control system **114** may include a pressure/flow controller **118** configured to control pressure and/or flow of supply air to evaporator **110**. In some examples, pressure/flow controller **118** may be configured to control air supply system **124** to generate a particular flow of supply air from air supply system **124**. For example, pressure/flow controller **118** may be configured to send control signals to air supply system **124** to control a pressure of supply air, such as by controlling an amount of bleed air or a speed of a load compressor (via an APU) or cabin air compressor. In some examples, pressure/flow controller **118** may be configured to control ram air system **102** in order to receive a particular flow of supply air from ram air system **102**.

[0032] Vapor cooling system (e.g., vapor compression systems) discussed herein may include a centrifugal compressor to compress a vapor refrigerant and cool various components of the compressor using the vapor refrigerant as a cooling medium. For example, a compressor motor, shaft bearings, and other components of the centrifugal compressor that receive power and/or experience friction may produce heat that needs to be removed. Rather than remove this heat using external cooling, which may be heavy or inefficient to operate, centrifugal compressors described herein may use the incoming vapor refrigerant to cool various components of the centrifugal compressor, thereby reducing or eliminating external cooling of the centrifugal compressor.

[0033] Centrifugal compressors described herein may be configured to form a relatively tight containment to hermetically seal the vapor refrigerant within the vapor compression system. FIG. 2 is a conceptual diagram illustrating an example VCC **112**. VCC **112** may include a first-stage compressor wheel **202A** and a second-stage compressor wheel **202B** (collectively referred to herein as “compressor wheels **202B**”). VCC **112** may extend along a longitudinal axis **204** and may include compressor wheels **202** disposed along longitudinal axis **204** and within compressor housing **203** of VCC **112**. Each of compressor wheels **202** may be a centrifugal compressor. VCC **112** may direct refrigerant **205** from inlet port **206** of housing **203** through compressor wheels **202** (e.g., through first-stage compressor wheel **202A** and second-stage compressor wheel **202B** in series) and out of one or more outlet port(s) **221** of housing **203**. Refrigerant **205** may enter VCC **112** at inlet port **206** from evaporator **110** of VCS **104** and exit VCC **112** at outlet port(s) **221** into condenser **106** of VCS **104**. While FIG. 2 illustrates two compressor wheels **202**, other example VCC **112** described herein may include three or more compressor wheels **202** coupled in series.

[0034] Each compressor wheel **202** may include a compressor body extending along and around longitudinal axis **204**. The compressor body may define a cylindrical body around longitudinal axis **204**. First-stage compressor wheel **202A** may be more distal to second-stage compressor wheel **202B** (i.e., closer to inlet port **206** than second-stage compressor wheel **202B**). Housing **203** may be configured to define a first chamber **212A** configured to retain first-stage compressor wheel **202A** and a second chamber **202B** configured to retain second-stage compressor wheel **202B**. First chamber **212A** may be fluidically connected to second chamber **212B** via one or more pathways **215** within housing **203**. In such examples, refrigerant **205** may exit the outlet of first-stage compressor wheel **202A**, enter first diffuser **214**, and enter the inlet of second-stage compressor

wheel **202B** through pathway(s) **215**. Refrigerant **205** may exit the outlet of second-stage compressor wheel **202B**, pass through second diffuser **218**, and exit VCC **112** through outlet port(s) **221**.

[0035] One or more seals **216** may be disposed at or around an inlet of each of compressor wheels **202**. Seals **216** may include one or more types of seals used in compressor systems including, but are not limited to, carbon seals or labyrinth seals. Seals **216** may interface with an outer surface of compressor wheels **202** to inhibit flow of refrigerant **205** around compressor wheels **202** and into one or more of chambers **212** (e.g., from inlet port **206** directly into first chamber **212A** and bypassing first-stage compressor wheel **202A**, from pathway(s) **215** directly into second chamber **212B** and bypassing second-stage compressor wheel **202B**).

[0036] Each compressor wheel **202** may include a corresponding impeller **210A**, **210B** (collectively referred to herein as “impellers **210**”) within a corresponding compressor body. Each of impellers **210** may include one or more blades disposed within an inner chamber of the compressor body. The inner chamber may be configured to retain refrigerant **205** within the compressor body. The one or more blades may be disposed around the circumference of the inner chamber of the compressor body. The one or more blades may be configured to rotate about longitudinal axis **204** as the respective compressor wheel **202** rotates about longitudinal axis **204**. During rotation of compressor wheels **202**, the one or more blades may push refrigerant **205** within the inner chambers of compressor wheels **202**, thereby pressurizing refrigerant **205**.

[0037] Housing **203** may define a channel **211** along longitudinal axis **204**. Channel **211** may be configured to retain a drive shaft extending from a centrifugal motor. The drive shaft may include an inner shaft **208** disposed within an outer shaft **306**. Inner shaft **208** may be directly coupled to the motor and outer shaft **306** may be affixed to inner shaft **208**. Each of compressor wheels **202** may include features disposed within channel **211** or on each compressor body (e.g., on outer shaft **306**) which may removably or permanently affix compressor wheels **202** to inner shaft **208**. In some examples, as illustrated in FIG. 2, at least a portion of at least one compressor wheel **202** (e.g., of second-stage compressor wheel **202B**) may form outer shaft **306** around inner shaft **208**. Outer shaft **306** may be received within a recess **307** in first-stage compressor wheel **202A**, e.g., to affix second-stage compressor wheel **202B** to first-stage compressor wheel **202A**. When compressor wheels **202** are affixed to inner shaft **208**, rotation of inner shaft **208** about longitudinal axis **204** may cause compressor wheels **202** to rotate about longitudinal axis **204** in a same direction as inner shaft **208**, thereby causing impellers **210** within compressor wheels **202** to compress refrigerant **205**.

[0038] VCC **112** may include two or more helical seals **220A**, **220B** (collectively referred to herein as “helical seals **220**”) disposed within channel **211**. Helical seals **220** may be disposed on inner shaft **208**, e.g., directly on and/or integrated into inner shaft **208**. In some examples, helical seals **220** may be disposed on an outer surface of an outer shaft configured to receive inner shaft **208**). Helical seals **220** may define a helical structure revolving around the circumference of inner shaft **208** or the outer shaft. Each helical seal **220** may define at least one complete revolution around the circumference of inner shaft **208** or the outer shaft.

[0039] Helical seals **220** may define different directions of rotations. For each helical seal **220**, the direction of rotation may be the same as or different from the direction of rotation for compressor wheels **202** and/or inner shaft **208**. For example, as illustrated in FIG. 2, first helical seal **220A** is wound in a same direction as the direction of rotation for compressor wheels **202** and inner shaft **208** (e.g., clockwise) and second helical seal **220B** is wound in the opposite direction (e.g., counterclockwise) as the direction of rotation for compressor wheels **202** and inner shaft **208**. When inner shaft **208** rotates about longitudinal axis **204**, inner shaft **208** may cause helical seals **220** to rotate about longitudinal axis **204** within channel **211**. When rotated, helical seals **220** may urge refrigerant **205** around each helical seal **220** to travel in a particular direction (e.g., through the respective helical seal **220**) based on the direction of winding of the respective helical seal **220**. For

example, as illustrated in FIG. 2, when rotated, first helical seal **220A** is configured to urge refrigerant **205** distally towards first-stage compressor wheel **202A** and second helical seal **220B** is configured to urge refrigerant **205** proximally towards second-stage compressor wheel **202B**. In such examples, helical seals **220** may urge refrigerant **205** back towards a closest point of entry into channel **211** and inhibit flow of refrigerant **205** across the entire longitudinal length of channel **211**. [0040] Each compressor wheel **202** may define an inlet and an outlet. For each compressor wheel **202**, the inlet and the outlet may each extend at least partially around longitudinal axis **204**. For example, for each compressor wheel **202**, the inlet and the outlet may define a complete ring around longitudinal axis **204**.

[0041] Housing **203** may define one or more pathways **215** fluidically connecting compressor wheels **202**. Pathway(s) **215** may be fully enclosed and may revolve at least partially around longitudinal axis **204**. For example, as illustrated in FIG. 2, pathway(s) **215** may define a torus shape radially outwards of compressor wheels **202**. Outlet port(s) **221** may be configured to output refrigerant **205** out of VCC **112**. Outlet port(s) **221** may define one or more openings fluidically coupled to a downstream component within the refrigeration circuit (e.g., condenser **106**). In some examples, as illustrated in FIG. 2, outlet port(s) **221** defines a torus shape radially outwards of compressor wheels **202**. Housing **203** may hermetically seal refrigerant **205** from an environment outside of VCC **112** aside from inlet port **206** and outlet port(s) **221**.

[0042] Refrigerant **205** may include a low-pressure refrigerant (e.g., a refrigerant having a relatively low saturation vapor pressure). For example, the refrigerant gas may be R-1233zd, r236fa, or r245fa, or a similar low-pressure refrigerant, as described herein.

[0043] A compressor motor may be coupled to inner shaft **208** and may be configured to rotate inner shaft **208** to rotate compressor wheels **202**. The centrifugal motor may include windings of a stator coupled to compressor wheels **202** and/or housing **203** and a rotor coupled to inner shaft **208**. Windings of the stator may be configured to receive an electrical signal from motor controller **120** and generate a dynamic magnetic field to drive the rotor. In some examples, the compressor motor may be configured to rotate compressor wheels **202** at about 80,000 rotations per minute (RPM) or greater. For example, the compressor motor may be configured to spin compressor wheels **202** at approximately 120,000 RPM.

[0044] FIG. 3 is a conceptual diagram illustrating an example second-stage compressor wheel **202B** and example inner shaft **208** of the example VCC **112** of FIG. 2. Second-stage compressor wheel **202B** may extend from a first end **302A** to a second end **302B** along longitudinal axis **204**. In some examples, as illustrated in FIG. 3, inner shaft **208** may be disposed within and coupled to an outer shaft **306**. Outer shaft **306** may be integrated into second-stage compressor wheel **202B**, as illustrated in FIG. 3. In some examples, outer shaft **306** is a separate component from second-stage compressor wheel **202B** and may be removably or fixedly attached to second-stage compressor wheel **202B**.

[0045] Second-stage compressor wheel **202B** may include compressor body **304** extending from second end **302B** towards first end **302A** and defining an inner chamber of second-stage compressor wheel **202B**. Compressor body **304** may define an inlet **303** and an outlet **305** of second-stage compressor wheel **202B**. The inner chamber of second-stage compressor wheel **202B** may extend from inlet **303** to outlet **305** and may include impeller **210B** (pictured in FIG. 2) disposed within the inner chamber.

[0046] Outer shaft **306** may define an elongated cylindrical body extending along longitudinal axis **204**. Outer shaft **306** may define an inner lumen sized to receive inner shaft **208**. In some examples, outer shaft **306** may include one or more attachment features disposed on an inner surface defining the inner lumen. The one or more attachment features may interface with corresponding features on the outer surface of inner shaft **208** to affix outer shaft **306** to inner shaft **208**. In some examples, inner shaft **208** is affixed to second-stage compressor wheel **202B** and, by extension, outer shaft **306** at a location within compressor body **304**.

[0047] Helical seals **220** may be affixed to outer shaft **306** (e.g., as illustrated in FIG. 3) or to an outer surface of inner shaft **208** (e.g., in the event that no outer shaft **306** is present). As illustrated in FIG. 3, helical seals **220** are wound in different directions and are defined by different parameters. For example, different helical seals **220** may define different pitches, number of complete revolutions, pitch angles, height (e.g., along a reference axis orthogonal to longitudinal axis **204**), or the like.

[0048] In the example illustrated in FIG. 3, first helical seal **220A** is wound about outer shaft **306** in a same direction as the direction of revolution of inner shaft **208** while second helical seal **220B** is wound about outer shaft **306** in the opposite direction. Additionally, in the example illustrated in FIG. 3, first helical seal **220A** extends towards first end **302A** while second helical seal **220B** extends towards second end **302B**. In such examples, first helical seal **220A** may direct refrigerant **205** around outer shaft **306** and/or inner shaft **208** distally towards first-stage compressor wheel **202A** while second helical seal **220B** may direct refrigerant **205** around outer shaft **306** and/or inner shaft **208** proximally towards second-stage compressor wheel **202B**, e.g., towards inlet **303**. For example, second helical seal **220B** may block and re-direct distal flow of refrigerant **205** (e.g., from inlet **303**) back towards inlet **303**. Additionally, the direction of winding of second helical seal **220B**, (e.g., around longitudinal axis **204**, along longitudinal axis **204**) may help urge refrigerant **205** located around outer shaft **306** and/or inner shaft **208** at a position distal to the position of second helical seal **220B** back towards inlet **303** of second compressor wheel **202B**. Helical seals **220** may be longitudinally adjacent to each or may be separated by a set distance along the longitudinal length of outer shaft **306**. In some examples, helical seals **220** may be positioned along the longitudinal length of outer shaft **306**, e.g., to be evenly distributed along the length of outer shaft **306**. The even distribution may facilitate balancing of outer shaft **306** and inner shaft **208** within channel **211**.

[0049] Compressor wheels **202** (e.g., compressor wheels **202A**, **202B**) may rotate about longitudinal axis **204** in a direction **308**. Direction **308** may be clockwise (e.g., as illustrated in FIG. 3) or counterclockwise. Inner shaft **208** may rotate about longitudinal axis **204** in direction **308** and cause compressor wheels **202A** (not pictured in FIGS. 3) and **202B** to rotate about longitudinal axis **204** in direction **308**.

[0050] Each helical seals **220** may be wound around longitudinal axis **204** in a corresponding direction (e.g., of direction **310A**, **310B**, collectively referred to herein as “directions **310**”). For example, as illustrated in FIG. 3, first helical seal **220A** may be wound around longitudinal axis **204** in first direction **310A** and second helical seal **220B** may be wound around longitudinal axis **204** in second direction **310B**. First direction **310A** may be the same as or may be different to (e.g., opposite to) second direction **310B**. In some examples, as illustrated in FIG. 3, first direction **310A** may be in the same direction as direction **308** and second direction **310B** may be opposite to direction **308**. In some examples, the winding of second helical seal **220B** in second direction **310B** may cause second helical seal **220B** to direct refrigerant **205** back towards second-stage compressor wheel **202B** when compressor wheels **202** and helical seals **220** are rotated about longitudinal axis **204** in direction **308**. In some examples, the winding of first helical seal **220A** in first direction **310A** may cause first helical seal **220A** to direct refrigerant **205** towards first-stage compressor wheel **202A** when compressor wheels **202** and helical seals **220** are rotated about longitudinal axis **204** in direction **308**.

[0051] FIG. 4A is a conceptual diagram illustrating an example of outer shaft **306** of FIG. 3 disposed within channel **211**. When outer shaft **306** is disposed within channel **211**, the longitudinal axes of outer shaft **306** and channel **211** may be aligned, e.g., along longitudinal axis **204**. Outer shaft **306** may define an outer diameter **404** and channel **211** may define an inner diameter **402**. An outer surface **410** of outer shaft **306** may be separated from an inner surface **408** of channel **211** by up to distance **406**. Outer shaft **306** may include a plurality of helical seals **220** extend away from outer surface **410** of outer shaft **306**. Each of helical seals **220** may extend up to a distance **412**

from outer surface **410**. A clearance between a maximum diameter of outer shaft **306** (e.g., as defined by the radially-outward most points on helical seals **220**) and inner surface **408** of channel **211** is distance **414**. For example, distance **406** from outer surface **410** to inner surface **408** is equal to a sum of distance **412** and distance **414**. Distance **414** may be up to about 0.5 millimeters (mm) (e.g., up to about 0.04 inches (in)).

[0052] Each of helical seals **220** may define same or different parameters as another of helical seals **220**. Parameters may include, but are not limited to, an outer diameter of each helical seal **220**, a pitch **416** of each helical seal **220**, a helical angle (e.g., helical angles **420A**, **420B**, collectively referred to herein as “helical angles **420**) of each helical seal **220**, a number of revolutions around the outer circumference of outer shaft **306** for each helical seal **220**, or the like. In some examples, where helical seals **220** include three or more helical seals, at least two helical seals **220** may define the same parameters.

[0053] Helical seal **220A** may define a same or different pitch **416** as helical seal **220B**. Each helical seal **220** may define a uniform or variable pitch along the longitudinal length of helical seal **220**. In some examples, helical seals **220** may define helical angles **420A**, **420B** with a same or different magnitude. In some examples, as illustrated in FIG. 4, helical seals **220A**, **220B** define respective helical angles **420A**, **420B** with the same magnitudes and different directions, such that helical seal **220A** extends towards one end (e.g., a distal end) of channel **211** and helical seals **220B** extends towards an opposite end (e.g., a proximal end) of channel **211**. Helical seals **220** may each define at least two complete revolutions around the circumference of outer shaft **306** (e.g., at least three complete revolutions. The number of complete revolutions may vary as a function of outer diameter **404**. For example, an increased outer diameter **404** may require helical seals **220** to complete more revolutions around the circumference of outer shaft **306** (e.g., to achieve substantially similar sealing properties), or vice versa.

[0054] In some examples, helical seals **220** may be longitudinally adjacent to each other. In some examples, helical seals **220** may be separated by up to distance **418**. Outer shaft **306** may include one or more other seals disposed between helical seals **220**. The one or more other seals may include, but are not limited to, carbon seals, ring seals, or other helical seals. Helical seals **220** may be integral to or integrated into a drive shaft of VCC **112** (e.g., into outer shaft **306** and/or inner shaft **208**).

[0055] FIG. 4B is a conceptual diagram illustrating another view of outer shaft **306** within channel **211**. As illustrated in FIG. 4B, helical seals **220** may rotate with outer shaft **306** to inhibit flow of refrigerant **205** across helical seals **220** and re-direct the flow of refrigerant **205** back towards the entrances to channel **211** at the proximal and distal end of channel **211**. Refrigerant **205** may flow through helical seals **220** along pathways **422** to be expelled towards the entrances to channel **211**. First helical seal **220A** may be wound in a same direction as the direction of rotation of outer shaft **306** to force refrigerant **205** within channel **211** to travel along pathways **422** towards a distal entrance to channel **211** (e.g., at a distal end of channel **211**). Second helical seal **220B** may be wound in an opposite direction as the direction of rotation of outer shaft **306** to force refrigerant **205** within channel **211** to travel along pathways **422** towards a proximal entrance to channel **211** (e.g., at a proximal end of channel **211**).

[0056] Outer shaft **306** may include one or more features **424** extending through at least partially through outer shaft **306**. In some examples feature(s) **424** may include openings configured to retain a fixation feature coupling outer shaft **306** to inner shaft **208** disposed within outer shaft **306**. In some examples, feature(s) **424** may include openings, recesses, protrusions or the like to facilitate balancing of outer shaft **306** and inner shaft **208** within channel **211**. In some examples, feature(s) **424** may include one or more seals to further inhibit the flow of refrigerant **205** through channel **211**. Feature(s) **424** may be disposed between two longitudinally adjacent helical seals **220** (e.g., between helical seals **220A**, **220B**), distal to one or more helical seals **220** (e.g., distal to first helical seal **220A**), proximal to one or more helical seals **220** (e.g., proximal to second helical seal

220B), or at any other location along outer shaft **306**.

[0057] FIG. 5 is a flow diagram illustrating an example process of assembling VCC **112** of FIG. 2. While FIG. 5 illustrates the steps of the example process in one order, the steps may be performed by a manufacturing assembly in any other order. Additionally, while FIG. 5 is primarily described with reference to VCC **112** illustrated in FIGS. 2-4B, the example process may be performed by a manufacturing assembly to assemble any other VCC **112** described herein.

[0058] The manufacturing assembly may position first-stage compressor wheel **202A** within housing **203** (**502**). Housing **203** may define a first chamber **212A** configured to receive first-stage compressor wheel **202A**. The manufacturing assembly may position first-stage compressor wheel **202A** within first chamber **212A** such that an inner lumen within first-stage compressor wheel **202A** configured to receive inner shaft **208** is aligned with longitudinal axis **204**. When first-stage compressor wheel **202A** is positioned within first chamber **212A**, an inlet of first-stage compressor wheel **202A** is orientated towards inlet port **206** of VCC **112**. When first-stage compressor wheel **202A** is positioned within first chamber **212A**, the outlets of first-stage compressor wheel **202A** may be aligned with first diffuser **214** and the entrances to pathway(s) **215**.

[0059] The manufacturing assembly may connect a second-stage compressor wheel **202B** to first-stage compressor wheel **202A** by at least partially inserting an outer shaft **306** of the drive shaft into first-stage compressor wheel **202A** (**504**). Outer shaft **306** may be integral or otherwise fixedly attached to second-stage compressor wheel **202B** and may extend distally from a distal end of second-stage compressor wheel **202B**. The manufacturing assembly may assemble one or more portions of housing **203** around a proximal end first-stage compressor wheel **202A** to retain first-stage compressor wheel **202A** within first chamber **212A** and define channel **211** extending to second chamber **212B**. The manufacturing assembly may insert second-stage compressor wheel **202B** into second chamber **212B** and outer shaft **306** into channel **211**. The manufacturing assembly may advance outer shaft **306** along longitudinal axis **204** and through channel **211** until a distal end of outer shaft **306** enters recess **307** at proximal end of first-stage compressor wheel **202A**. The manufacturing assembly may affix the distal end of outer shaft **306** to recess **307** of first-stage compressor wheel **202A** (e.g., via a friction fit, an adhesive, or one or more fixation features). When outer shaft **306** is disposed within channel **211**, helical seals **220** may be disposed within channel **211**.

[0060] The manufacturing assembly may insert inner shaft **208** of the drive shaft through an inner lumen of outer shaft **306** along longitudinal axis **204** (**506**). The inner lumen of outer shaft **306** may be sized to retain inner shaft **208**. The manufacturing assembly may insert inner shaft **208** through the inner lumens of second-stage compressor wheel **202B**, outer shaft **306**, and first-stage compressor wheel **202A**, e.g., until a distal end of inner shaft **208** is within or distal to the compressor body of first-stage compressor wheel **202A**. The manufacturing assembly may then affix inner shaft **208** to first-stage compressor wheel **202A** and to one or more of second-stage compressor wheel **202B** or outer shaft **306**. Once affixed, rotation of inner shaft **208** about longitudinal axis **204** may cause compressor wheels **202** and outer shaft **306** to rotate about longitudinal axis **204** in a same direction. In some examples, inner shaft **208** may be inserted into outer shaft **306** and/or affixed to second-stage compressor wheel **202B** and/or to outer shaft **306** prior to the insertion of outer shaft **306** into recess **307**.

[0061] Once the manufacturing assembly affixes inner shaft **208** to first-stage compressor wheel **202A**, second-stage compressor wheel **202B**, and/or to outer shaft **306**, the manufacturing assembly may affix a proximal end of inner shaft **208** to a centrifugal motor of VCC **112**. In response to a torque from the motor, inner shaft **208** may transmit the received torque to compressor wheels **202** and outer shaft **306** to cause compressor wheels **202** and outer shaft **306** to rotate about longitudinal axis **204**, e.g., to pressurize and output pressurized refrigerant **205** through VCC **112**. When outer shaft **306** is rotating within channel **211**, helical seals **220** may inhibit and re-direct refrigerant **205** within channel **211** to prevent the unintended flow of refrigerant **205** between compressor wheels

202 through channel **211**. The manufacturing assembly may then complete assembly of the remainder of VCC **112** (e.g., the remainder of housing **203**) to fully enclose and encapsulate compressor wheels **202** within housing **203**.

[0062] The following examples may illustrate one or more of the techniques of this disclosure.

[0063] Example 1: a compressor comprising: an drive shaft; a first compressor wheel; a second compressor wheel connected to the first compressor wheel via the drive shaft; and two or more helical seals disposed on the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the drive shaft and around a circumference of the drive shaft, wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel.

[0064] Example 2: the compressor of example 1, wherein the two or more helical seals comprise a first helical seal adjacent to the first compressor wheel and a second helical seal adjacent to the second compressor wheel.

[0065] Example 3: the compressor of example 2, wherein the compressor is configured to direct the fluid from the first compressor wheel to the second compressor wheel, wherein the first helical seal is wound in a same direction as a direction of rotation of the drive shaft, and wherein the second helical seal is wound in an opposite direction relative to the direction of rotation of the drive shaft.

[0066] Example 4: the compressor of example 3, wherein the first helical seal is configured to direct the fluid around the drive shaft towards the first compressor wheel, and wherein the second helical seal is configured to direct the fluid around the drive shaft towards the second compressor wheel.

[0067] Example 5: the compressor of any of examples 1-4, wherein the two or more helical seals are integrated into the drive shaft.

[0068] Example 6: the compressor of any of examples 1-5, wherein the first compressor wheel extends along a longitudinal axis from a first end to a second end, wherein the second compressor wheel extends along the longitudinal axis from a third end to a fourth end, and wherein the two or more helical seals are longitudinally disposed between the second end of the first compressor wheel and the third end of the second compressor wheel.

[0069] Example 7: the compressor of example 6, wherein the drive shaft comprises: an outer shaft extending from the first compressor wheel to the second compressor wheel; and an inner shaft disposed radially inwards of the outer shaft and extending through the first compressor wheel and the second compressor wheel, wherein the two or more helical seals are disposed on the outer shaft of the drive shaft.

[0070] Example 8: the compressor of any of examples 1-7, wherein each helical seal of the two or more helical seals defines at least two complete revolutions around the circumference of the drive shaft.

[0071] Example 9: the compressor of any of examples 1-8, wherein each helical seal of the two or more helical seals extends from the outer surface of the drive shaft to an outermost edge.

[0072] Example 10: the compressor of any of examples 1-9, further comprising a housing configured to retain the first compressor wheel, the second compressor wheel, and the drive shaft, wherein the housing defines a channel sized to retain the drive shaft, and wherein a distance between the outermost edge of each helical seal and an inner surface of the channel is up to 0.05 millimeters (mm).

[0073] Example 11: the compressor of any of examples 1-10, further comprising one or more secondary seals disposed on the outer surface of the drive shaft.

[0074] Example 12: the compressor of any of examples 1-11, wherein for each helical seal of the two or more helical seals, a first set of one or more parameters defining the helical seal are substantially similar to a second set of one or more parameters defining another helical seal of the two or more helical seals, and wherein each of the first set of one or more parameters and the second set of one or more parameters comprises one or more of: a pitch of the helical seal; a helical

angle of the helical seal; a number of revolutions around the circumference of the drive shaft; a longitudinal length of the helical seal; or an outer diameter of the helical seal.

[0075] Example 13: the compressor of example 12, wherein each helical seal of the two or more helical seals defines a same number of revolutions around the circumference of the drive shaft.

[0076] Example 14: a compressor system comprising: a compressor comprising: a first compressor wheel; a second compressor wheel; an drive shaft extending along a longitudinal axis from a proximal end to a distal end, wherein the drive shaft extends through the first compressor wheel and the second compressor wheel and connects the first compressor wheel to the second compressor wheel, and wherein the distal end of the drive shaft is distal to the first compressor wheel; and two or more helical seals disposed on the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the drive shaft and around a circumference of the drive shaft; a housing configured to retain the compressor, the housing comprising: a channel extending along the longitudinal axis and configured to retain the drive shaft, wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel through the channel; and a motor coupled to the proximal end of the drive shaft and configured to apply a rotary force on the drive shaft.

[0077] Example 15: the compressor system of example 14, wherein the two or more seals comprise a first helical seal adjacent to the first compressor wheel and a second helical seal adjacent to the second compressor wheel.

[0078] Example 16: the compressor system of example 15, wherein the housing comprises one or more channels fluidically connecting the first compressor wheel to the second compressor wheel, wherein rotation of the drive shaft in a first direction causes the first compressor wheel to direct the fluid along the one or more channels and into the second compressor wheel.

[0079] Example 17: the compressor system of example 16, wherein the first helical seal is wound in the first direction, and wherein the second helical seal is wound in a second direction opposite the first direction.

[0080] Example 18: the compressor system of any of examples 15-17, wherein the first helical seal is configured to direct fluid entering a distal end of the channel towards the first compressor wheel, and wherein the second helical seal is configured to direct fluid entering a proximal end of the channel towards the second compressor wheel.

[0081] Example 19: the compressor system of any of examples 14-18, wherein the two or more helical seals are integrated into the drive shaft.

[0082] Example 20: the compressor system of any of examples 14-19, wherein each helical seal of the two or more helical seals defines at least two complete revolutions around the circumference of the drive shaft.

[0083] Example 21: the compressor system of any of examples 14-20, wherein a distance between an inner surface of the channel and an outermost edge of at least one helical seal of the two or more helical seals is up to 0.05 millimeters (mm).

[0084] Example 22: the compressor system of any of examples 14-21, wherein the compressor further comprises one or more secondary seals disposed on the outer surface of the drive shaft.

[0085] Example 23: the compressor system of any of examples 14-22, wherein one or more parameters for each helical seal of the two or more helical seals are substantially similar to the one or more parameters for another helical seal of the two or more helical seals, and wherein for each helical seal of the two or more helical seals, the one or more parameters comprises one or more of: a pitch of the helical seal; a helical angle of the helical seal; a number of revolutions completed by the helical seal around the circumference of the drive shaft; a longitudinal length of the helical seal along the longitudinal axis; or an outer diameter of the helical seal.

[0086] Example 24: the compressor system of example 23, wherein each helical seal of the two or more helical seals completes a same number of revolutions around the circumference of the drive shaft.

[0087] Example 25: a method comprising: position a first compressor wheel of a compressor within a housing; connecting a second compressor wheel of the compressor to the first compressor wheel by at least partially inserting an outer shaft of an drive shaft of the compressor into the first compressor wheel, wherein the outer shaft is connected to the second compressor wheel, and wherein the outer shaft of the drive shaft comprises: two or more helical seals disposed on the outer shaft, each helical seal of the two or more helical seals extending away from an outer surface of the outer shaft and around a circumference of the outer shaft; inserting an inner shaft of the drive shaft through an inner lumen of the outer shaft along a longitudinal axis of the compressor; and affixing the inner shaft to the first compressor wheel and the second compressor wheel.

[0088] Example 26: the method of example 25, wherein the housing defines an inner channel extending along the longitudinal axis from a proximal end to a distal end, and wherein inserting the outer shaft into the first compressor wheel comprises: inserting the outer shaft into the proximal end of the inner channel; advancing the outer shaft along the longitudinal axis until a distal end of the outer shaft extends past the distal end of the inner channel; and affixing the distal end of the outer shaft to the first compressor wheel, wherein when the outer shaft is affixed to the first compressor wheel, the two or more helical seals are positioned within the inner channel.

[0089] Example 27: the method of example 26, wherein a distance between an outermost edge of each helical seal of the two or more helical seals and an inner surface of the inner channel is up to 0.05 millimeters (mm).

[0090] Example 28: the method of any of examples 26 or 27, wherein the two or more helical seals comprises a first helical seal adjacent to the distal end of the outer shaft and a second helical seal adjacent to the second compressor wheel, wherein the first helical seal is configured to direct fluid entering the inner channel from the distal end of the inner channel towards the first compressor wheel, and wherein the second helical seal is configured to direct fluid entering the inner channel from the proximal end of the inner channel towards the second compressor wheel.

[0091] Example 29: the method of any of examples 25-28, wherein the inner shaft is coupled to a motor, wherein the motor is configured to cause the compressor to rotate about the longitudinal axis in a first direction, and wherein the first helical seal is wound around the circumference of the outer shaft in the first direction and the second helical seal is wound around the circumference of the outer shaft in a second direction opposite the first direction.

[0092] Example 30: the method of any of examples 25-29, wherein the two or more helical seals are integrated into the outer shaft of the drive shaft.

[0093] Example 31: the method of any of examples 25-30, wherein each helical seal of the two or more helical seals defines at least two complete revolutions around the circumference of the outer shaft.

[0094] Example 32: the method of any of examples 25-31, wherein one or more parameters for each helical seal of the two or more helical seals are substantially similar to the one or more parameters for another helical seal of the two or more helical seals, and wherein for each helical seal of the two or more helical seals, the one or more parameters comprises one or more of: a pitch of the helical seal; a helical angle of the helical seal; a number of revolutions completed by the helical seal around the circumference of the outer shaft of the drive shaft; a longitudinal length of the helical seal along the longitudinal axis; or an outer diameter of the helical seal.

[0095] A “vehicle” may be an aircraft, a land vehicle such as an automobile, or a water vehicle such as a ship or a submarine. An “aircraft” as described and claimed herein may include any fixed-wing or rotary-wing aircraft, airship (e.g., dirigible or blimp buoyed by helium or other lighter-than-air gas), suborbital spaceplane, spacecraft, expendable or reusable launch vehicle or launch vehicle stage, or other type of flying device. An “aircraft” as described and claimed herein may include any crewed or uncrewed craft (e.g., uncrewed aerial vehicle (UAV), flying robot, or automated cargo or parcel delivery drone or other craft).

Claims

1. A compressor comprising: a first compressor wheel; a second compressor wheel; a drive shaft connecting the first compressor wheel to the second compressor wheel, the drive shaft comprising: an outer shaft extending from the first compressor wheel to the second compressor wheel; and an inner shaft disposed radially inwards of the outer shaft and extending through the first compressor wheel and the second compressor wheel; and two or more helical seals disposed on the outer shaft of the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the outer shaft and around a circumference of the outer shaft, wherein an outer diameter of each helical seal of the two or more helical seals is greater than an outer diameter of the outer shaft, and wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel.
2. The compressor of claim 1, wherein the two or more helical seals comprise a first helical seal adjacent to the first compressor wheel and a second helical seal adjacent to the second compressor wheel.
3. The compressor of claim 2, wherein the compressor is configured to direct the fluid from the first compressor wheel to the second compressor wheel, wherein the first helical seal is wound in a same direction as a direction of rotation of the drive shaft, and wherein the second helical seal is wound in an opposite direction relative to the direction of rotation of the drive shaft.
4. The compressor of claim 3, wherein the first helical seal is configured to direct the fluid around the drive shaft towards the first compressor wheel, and wherein the second helical seal is configured to direct the fluid around the drive shaft towards the second compressor wheel.
5. The compressor of claim 1, wherein the two or more helical seals are integrated into the drive shaft.
6. The compressor of claim 1, wherein the first compressor wheel extends along a longitudinal axis from a first end to a second end, wherein the second compressor wheel extends along the longitudinal axis from a third end to a fourth end, and wherein the two or more helical seals are longitudinally disposed between the second end of the first compressor wheel and the third end of the second compressor wheel.
7. (canceled)
8. The compressor of claim 1, wherein each helical seal of the two or more helical seals defines at least two complete revolutions around the circumference of the drive shaft.
9. The compressor of claim 1, wherein each helical seal of the two or more helical seals extends from the outer surface of the drive shaft to an outermost edge.
10. The compressor of claim 1, further comprising a housing configured to retain the first compressor wheel, the second compressor wheel, and the drive shaft, wherein the housing defines a channel sized to retain the drive shaft, and wherein a distance between the outermost edge of each helical seal and an inner surface of the channel is up to 0.05 millimeters (mm).
11. The compressor of claim 1, wherein for each helical seal of the two or more helical seals, a first set of one or more parameters defining the helical seal are substantially similar to a second set of one or more parameters defining another helical seal of the two or more helical seals, and wherein each of the first set of one or more parameters and the second set of one or more parameters comprises one or more of: a pitch of the helical seal; a helical angle of the helical seal; a number of revolutions around the circumference of the drive shaft; a longitudinal length of the helical seal; or an outer diameter of the helical seal.
12. The compressor of claim 11, wherein each helical seal of the two or more helical seals defines a same number of revolutions around the circumference of the drive shaft.
13. A compressor system comprising: a compressor comprising: a first compressor wheel; a second compressor wheel; a drive shaft extending along a longitudinal axis from a proximal end to a distal

end, the drive shaft comprising: an outer shaft extending from the first compressor wheel to the second compressor wheel; and an inner shaft disposed radially inwards of the outer shaft and extending through the first compressor wheel and the second compressor wheel, wherein the drive shaft connects the first compressor wheel to the second compressor wheel, and wherein the distal end of the drive shaft is distal to the first compressor wheel; and two or more helical seals disposed on the outer shaft of the drive shaft, each helical seal of the two or more helical seals extending away from an outer surface of the outer shaft and around a circumference of the outer shaft, wherein an outer diameter of each helical seal of the two or more helical seals is greater than an outer diameter of the outer shaft; a housing configured to retain the compressor, the housing comprising: a channel extending along the longitudinal axis and configured to retain the drive shaft, wherein the two or more helical seals are configured to inhibit flow of fluid between the first compressor wheel and the second compressor wheel through the channel; and a motor coupled to the proximal end of the drive shaft and configured to apply a rotary force on the drive shaft.

14. The compressor system of claim 13, wherein the two or more seals comprise a first helical seal adjacent to the first compressor wheel and a second helical seal adjacent to the second compressor wheel.

15. The compressor system of claim 14, wherein the housing comprises one or more channels fluidically connecting the first compressor wheel to the second compressor wheel, wherein rotation of the drive shaft in a first direction causes the first compressor wheel to direct the fluid along the one or more channels and into the second compressor wheel.

16. The compressor system of claim 15, wherein the first helical seal is wound in the first direction, and wherein the second helical seal is wound in a second direction opposite the first direction.

17. The compressor system of claim 14, wherein the first helical seal is configured to direct fluid entering a distal end of the channel towards the first compressor wheel, and wherein the second helical seal is configured to direct fluid entering a proximal end of the channel towards the second compressor wheel.

18. The compressor system of claim 13, wherein the two or more helical seals are integrated into the drive shaft.

19. A method comprising: position a first compressor wheel of a compressor within a housing; connecting a second compressor wheel of the compressor to the first compressor wheel by at least partially inserting an outer shaft of a drive shaft of the compressor into the first compressor wheel, wherein the outer shaft is connected to the second compressor wheel, and wherein the outer shaft of the drive shaft comprises: two or more helical seals disposed on the outer shaft, each helical seal of the two or more helical seals extending away from an outer surface of the outer shaft and around a circumference of the outer shaft; inserting an inner shaft of the drive shaft through an inner lumen of the outer shaft along a longitudinal axis of the compressor; and affixing the inner shaft to the first compressor wheel and the second compressor wheel.

20. The method of claim 19, wherein the housing defines an inner channel extending along the longitudinal axis from a proximal end to a distal end, and wherein inserting the outer shaft into the first compressor wheel comprises: inserting the outer shaft into the proximal end of the inner channel; advancing the outer shaft along the longitudinal axis until a distal end of the outer shaft extends past the distal end of the inner channel; and affixing the distal end of the outer shaft to the first compressor wheel, wherein when the outer shaft is affixed to the first compressor wheel, the two or more helical seals are positioned within the inner channel.

21. The compressor of claim 1, wherein at least a portion of the second compressor wheel forms the outer shaft of the drive shaft.
